Rack-Level Micro-Climate and Growth Optimizer for Oyster mushroom Farms

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Project Proposal Report

Dissanayake R.M.T.S (IT22062406)

B.Sc. (Hons) Degree in Information Technology - Specializing in Information Systems Engineering

Department of Computer System Engineering

Faculty of Computing

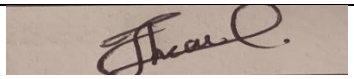
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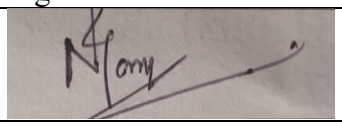
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Dissanayake R.M.T. S	IT22062406	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Supervisor	Ms. Narmada Gamage	

ABSTRACT

Oyster mushroom cultivation relies on an appropriate balance of temperature and humidity. A slight change in either of these can affect the growth rate, yield, or quality of the crop. Farmers are left to face moisture stress, reduced yield, or increased expenditure. Though many devices are available to monitor the environment, many are only capable of monitoring humidity or have no predictive capabilities or ability to interpret the data to take appropriate action.

The need for an alternative was identified, hence the need for a predictive moisture management system based on the Transpiration Stress Index (TSI) calculated using temperature and humidity data. The system continuously monitors the environment using sensors, calculates the TSI using a specific algorithm, and initiates appropriate action as soon as the moisture stress is sensed. The system is designed using IoT-compatible hardware, ESP32 microcontrollers, cloud storage, and an interface for easy monitoring.

The expected outcome is to have real-time monitoring of the environment for cultivation, timely sensing of moisture stress, better growth of mushrooms, and increased yield. The system is expected to minimize crop loss for farmers and enable them to take data-driven decisions.

Impact on Society: The system can be used for small-scale cultivation of mushrooms by farmers to adopt smart farming practices. The system can be used for commercial-scale production of mushrooms.

Keywords: Oyster mushroom, Transpiration Stress Index, Predictive moisture management, IoT, Environmental monitoring

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
TSI	Transpiration Stress Index
IoT	Internet of Things
RH	Relative Humidity
VPD	Vapor Pressure Deficit
Arduino	microcontroller platform
PCB	Printed Circuit Board
LED	Light Emitting Diode
MCU	Microcontroller Unit
Wi-Fi	Wireless Fidelity
API	Application Programming Interface
GPS	Global Positioning System
HVAC	Heating, Ventilation, and Air Conditioning
PM	Particulate Matter

1.INTRODUCTION

1.1.Background and Context

Oyster mushroom cultivation is one of the major agricultural activities that have gained popularity worldwide due to its ease of cultivation, minimal space requirements, and good income generation for small-scale farmers. The cultivation of oyster mushrooms is largely dependent on environmental factors, including temperature and humidity levels. The balance of these two factors is critical in ensuring that mushroom growth is healthy and that the yield is maximized. Failure to control these factors may result in moisture stress, slow growth, or even failure of crops.

Temperature and humidity are critical environmental parameters that affect mushroom cultivation. Research has indicated that the optimal temperature for growing oyster mushrooms is between 20 and 26°C, with humidity levels between 80% and 90%. In case these parameters are not observed, the growth process may be negatively affected, resulting in poor-quality mushrooms. In small mushroom cultivation farms, environmental monitoring is done through manual methods by observing changes in temperature and humidity levels, followed by sprinkling of water on the crops. This is not effective due to the dynamic nature of environmental parameters, which changes very quickly.

There are various technological innovations that have been introduced to ensure better environmental monitoring for the cultivation of mushrooms. For example, IoT-based monitoring systems have been introduced to monitor the environmental conditions, such as temperature and humidity. These monitoring systems can ensure better environmental stability to produce better quality mushrooms. However, the existing monitoring systems are mainly designed to monitor either humidity or temperature alone. The existing monitoring systems are not designed to monitor the effect of both humidity and temperature on transpiration stress or moisture loss in mushrooms.

Therefore, the present research aims to introduce a new system called Temperature Humidity Based Transpiration Stress Index (TSI) for predictive moisture management in the cultivation of oyster mushrooms. The proposed system aims to analyze the environmental conditions such as temperature and humidity, to ensure better decision making for the cultivation of mushrooms.

The proposed system is aligned with the Sustainable Development Goal 2: Zero Hunger, which aims to improve agricultural productivity for sustainable food production. The proposed system is also aligned with the need to introduce innovations in smart agriculture or digital farming, which is an emerging need for modern agricultural research clusters.

1.1.Literature Review

The environment affects plants in many ways, including how plants develop and the productivity of crops. Among the many factors that influence plants, the most important are probably temperature and relative humidity. Temperature affects many physiological activities in plants, including photosynthesis, respiration, enzymes, and nutrient uptake. According to research, the rate of development in plants is very sensitive to changes in temperature. This is because heat and cold directly affect the physiological activities in plants. Therefore, plants develop physiological stress when the temperature changes and moves out of the range that supports development. As a result, there is a need to regulate the temperature within the range that supports development in plants to enhance agricultural productivity [1].

Another important factor in plants is relative humidity. This refers to the proportion of water vapor in the air to the air's capacity to hold the vapor at a given temperature. This process affects the rate at which plants lose water through transpiration. This process occurs when there is an uptake of water from the roots to the leaves and out of the plants through the stomata. When humidity is low, the air has the capacity to take in more water vapor, thus increasing the rate at which plants lose water. On the other hand, when humidity is high, there is less difference in the rate at which plants lose water and the capacity of the air to take in the vapor. Therefore, humidity is very important in ensuring that plants do not suffer moisture stress.

Many authors have pointed out the need to consider the two elements jointly because you cannot consider the effect of temperature without the effect of humidity. For example, it has been established in earlier research on the water relations of plants that increased temperature coupled with reduced humidity increases the deficit of vapour pressure, which accelerates the rate of water loss from the plant. This may lead to moisture stress if the plant does not have adequate water. Hence, it is important to understand the effect of temperature coupled with humidity to understand the regulation of plant moisture.

There have been several studies on the effect of temperature and humidity on vegetable crops and horticultural plants. These studies provide valuable information on the effect of environmental factors on plant physiology. For example, there was a research on the effect of relative humidity on the tolerance of watermelon seedlings to low temperature. It was established that the seedlings grew in high humidity showed better water potential in the leaves compared to those grown in low humidity. Additionally, the high humidity helped to stabilize the cell membrane to ensure proper photosynthesis.

Similarly, research on cucumber plants grown in greenhouse environments has shown that unfavorable combinations of temperature and humidity can significantly affect plant growth and physiological performance. The study observed that low temperature combined with high humidity conditions reduced photosynthetic activity and affected Plant metabolism is affected when the environment gets tough. There are effects on the

amount of chlorophyll, antioxidant enzyme activity, and hormone levels of the plants under stress conditions due to temperature and humidity. This study implies that these stress conditions, which are a result of the environment, affect the growth of the plants, hence the yield. Further studies were conducted with cucumber seedlings, revealing the same effects of stress conditions, which affect the biosynthesis of chlorophyll, thereby limiting the yield of the plants.

In recent times, the development of new technology has provided a new way of monitoring the environment of the farms. Smart farming, together with the Internet of Things (IoT), has been used to monitor the environment of the farms, including the levels of temperature, humidity, soil moisture, and light intensity. This involves the use of sensors connected to the internet through a wireless network, enabling the collection of information concerning the environment, hence keeping a close eye on the environment of the plants.

Environmental monitoring is an important aspect of mushroom farming since the mushroom is highly sensitive to the surrounding environment. Oyster mushrooms, for instance, require specific temperature and humidity levels to support the growth of healthy mycelium. If the environmental factors fluctuate beyond the required limits, the mushroom growth may be affected. In order to address the problem, researchers have developed wireless monitoring tools to track temperature and humidity levels within the mushroom growing space. The tools assist the farmers to monitor the environmental changes to provide the optimal growing conditions.

Lately, the use of IoT technology has gained popularity among mushroom farmers. The technology involves the use of sensors connected to the microcontroller to store the data on the cloud. The farmers can access the data remotely to monitor the temperature and humidity levels. The technology is useful to stabilize the growing environment to improve the mushroom farming process.

In addition to the use of the monitoring system, other researchers have developed an automated system to control the mushroom growing environment. The system utilizes the microcontroller technology, specifically the Arduino board, to control the temperature and humidity levels. The system provides the optimal growing conditions by adjusting the temperature and humidity levels to improve the mushroom growth. The system minimizes the need to frequently check the mushroom-growing environment.

These monitoring and controlling systems often focus on temperature and humidity as separate entities. While useful, they provide no means of understanding the role of the combination of temperature and humidity on moisture stress within mushroom production. In fact, some of these systems may only monitor the environmental conditions without providing an understanding of how the environmental conditions affect the mushroom.

Previous research on plant life revealed that moisture stress is actually a result of the combined effect of temperature and humidity and not individually from temperature and humidity alone. However, only a few studies have attempted to integrate this perspective on the environment into mushroom farming. As such, mushroom farming is likely to face a challenge in understanding the role of the environment in mushroom growth.

Hence, it is possible to gain a better understanding of the environmental factors affecting mushroom farming by combining temperature and humidity data. An environmental indicator that is a result of combining temperature and humidity is likely to improve moisture stress in mushroom farming.

1.2.Research Gap

There have been a number of studies that have introduced environmental monitoring technology into mushroom farming. Most of the technology has been able to help the farmer regulate the temperature and humidity levels within the optimal range. It has improved the farmer's ability to recognize changes in the environment within the growing rooms. However, most of the technology has been able to measure the temperature and humidity levels individually or regulate the levels to a constant state.

Although the monitoring technology has been able to provide the farmer with valuable information regarding the environment, it has not provided the farmer with a complete understanding of the relationship between the temperature and humidity levels in the growing environment. In the case of mushroom farming, the environmental conditions do not exist independently. For instance, the temperature and humidity levels work together to regulate the moisture levels in the mushrooms.

Mushrooms are very sensitive to their environment, especially when it comes to their exposure to temperature and humidity. When these two factors are affected, the moisture level of the mushrooms can be affected in a significant way. For example, if the humidity level decreases while the temperature increases, the mushrooms will start to transpire more rapidly. This will, in turn, cause them to lose more moisture, which will affect their growth negatively.

Although the relationship between the environment and moisture stress is important, most of the available monitoring equipment is not able to assess the impact of the relationship between the two factors on the moisture stress of the mushrooms. Most of the equipment, both hardware and software, available in the market is used to observe the relationship between the two factors, but it does not go a step further to assess the impact of the relationship between the two factors on the moisture stress of the mushrooms. This means that the farmers will still find it hard to identify the moisture stress conditions before the conditions start to affect the growth of the mushrooms.

In order to address the said constraint, a more integrated perspective is required to consider the interaction between temperature, humidity, and moisture stress in mushroom growing rooms.

In this context, a Temperature Humidity Based Transpiration Stress Index (TSI) is introduced as a framework to evaluate moisture stress more precisely. The index integrates temperature and humidity into a comprehensive framework to consider environmental stress related to moisture stress.

The novelty of the system is its use of a predictive approach to manage moisture in oyster mushroom cultivation by combining temperature and humidity into a singular stress index. Using an index-based approach, the system is able to provide a better understanding of environmental conditions and their impact on mushroom growth, which could potentially be used to develop a more advanced environmental monitoring system and agricultural practices in mushroom cultivation.

Features	**Existing Research**	**Limitation**	**Proposed System**	**Reference**
Environmental Monitoring	Sensor-based systems used to monitor farm environments	Focus mainly on data collection	Use environmental data for stress analysis	[1]
Temperature Monitoring	Temperature sensors widely used in agriculture	Temperature analyzed separately	Combine temperature with humidity	[2]
Humidity Monitoring	Relative humidity monitoring used in crop systems	No interaction analysis with temperature	Integrated temperature–humidity evaluation	[2]
IoT Monitoring	IoT systems monitor environmental parameters in real time	Mainly provide monitoring dashboards	Use data for predictive analysis	[1], [7]
Transpiration Analysis	Some crop studies analyze transpiration behavior	Not applied to mushroom cultivation	Apply transpiration concept to mushrooms	[5]
Moisture Stress Monitoring	Studies focus on environmental monitoring	No direct measurement of moisture stress	Estimate moisture stress using TSI	[3], [4]

Features	Existing Research	Limitation	Proposed System	Reference
Integrated Environmental Model	Limited integrated models in agriculture	Temperature and humidity treated independently	Integrated Temperature-Humidity Stress Index	[3]
Predictive Environmental Management	Existing systems provide real-time monitoring	Lack of prediction capability	Predictive moisture management approach	Proposed Study

2.OBJECTIVE

2.1 Main Objective

To develop a predictive environmental monitoring system for oyster mushroom farming that can calculate the Transpiration Stress Index (TSI) based on temperature and humidity readings and alert or adjust the moisture level for increased yields and reduced losses in the crop.

2.2 Specific Objectives

1. Maintain a constant watch on the environmental conditions in the growing room by using reliable sensors that monitor temperature and humidity levels continuously.
2. Develop a Transpiration Stress Index (TSI) by designing an algorithm that converts temperature and humidity levels into a measurement of moisture stress on the mushrooms.
3. Develop a user-friendly monitoring interface that displays environmental data, stress levels, and any problems that may occur in the system, ensuring that the farmer is always aware of the status.
4. Implement a feature for automatic moisture management by integrating a misting or humidification system that activates when the TSI levels indicate a high level of stress on the mushrooms.
5. Test the effectiveness of the Transpiration Stress Index prediction mechanism and the automatic moisture management feature in different environmental conditions.
6. Ensure that data-driven decision-making is possible by storing historical environmental data for analysis.

3.METHODOLOGY

3.1 Research Approach

The research methodology used is Design Science Research (DSR). The DSR methodology is based on designing and developing a technological solution to solve a specific problem. The research aims to design a system that can predict the moisture stress during oyster mushroom cultivation based on various environmental conditions such as temperature and humidity. The proposed system collects data on various environmental conditions using sensors installed in the area where mushrooms are grown. The collected data is then used to calculate the Transpiration Stress Index (TSI) to identify the level of moisture stress that can arise during oyster mushroom cultivation. This data can then be used by farmers to understand various environmental conditions and take necessary measures to ensure proper conditions for oyster mushroom cultivation. The proposed research aims to design a system to solve a specific problem. The Agile methodology is used to design the proposed system. This methodology allows the proposed system to be developed step by step. Each step is referred to as a sprint. During each sprint, different parts of the proposed system are developed. Once a part is developed, it is tested. The proposed system is then tested to identify how effectively it can predict moisture stress during oyster mushroom cultivation.

3.2.Overall System Description

The Temperature Humidity Based Transpiration Stress Index Component is a modular system within an oyster mushroom farm management application. The primary role of the Temperature Humidity Based Transpiration Stress Index Component is to provide a system to monitor the environmental parameters and predict the level of moisture stress in oyster mushrooms.

Frontend (Mobile App): The application will be created with React Native, which is a framework that allows the creation of mobile applications with a user-friendly interface. The application will provide a user interface where a farmer can view the real-time data on the environmental parameters such as temperature and humidity, as well as the calculated values of the TSI.

Backend (Server): The backend is implemented using Node.js and Express.js to provide fast APIs to process the sensor data, compute the TSI, and return the results to the frontend app. The server receives real-time updates from IoT sensors and also stores environmental and index data for analysis.

Database: MongoDB stores the collected data from the sensors, calculated TSI values, historical environmental data, and user data. The flexible schema allows for the easy storage of the time series data from the sensors and can be utilized for the analysis of the growth patterns and stress in the future.

IoT Integration: Environmental data is obtained through IoT sensors placed within the mushroom farm. The sensors send real-time data regarding the temperature and humidity levels

through the **MQTT** protocol. This enables the system to continuously update the TSI values based on the changes in the environment.

Transpiration Stress Index (TSI) Calculation: The central component calculates the TSI based on the interaction of temperature and humidity, which reflects the moisture stress experienced by mushrooms. This predictive index assists the farmer in understanding the impact of the environment on the hydration of the mushrooms.

Integration and Alerts: The values and conditions of TSI values and environmental conditions are indicated in the app's dashboard. The system can send out alerts when the value of TSI reaches a critical threshold. This allows the farmer to proactively manage moisture. The data gathered over time can also be used to improve the farmer's environmental control strategies.

Performance Goals: The system strives to offer exact real-time calculations of TSI, timely updates within seconds of sensor data, and alerting that enables farmers to combat moisture stress, promote mushroom development, and boost yields.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system aims to monitor environmental conditions during oyster mushroom cultivation and predict moisture stress using the Transpiration Stress Index. The proposed system has various layers that work together to achieve a goal of collecting data, processing it, and making it more useful for farmers.

The first layer is the sensor layer, which includes temperature and humidity sensors placed inside the environment where mushrooms are grown. The sensors collect data on environmental conditions. The collected data is then sent to a microcontroller. The microcontroller is an intermediate device that collects data from sensors.

Next in the process is the sending of the collected data to the processing layer for analysis of the collected environmental data. In this step, the system computes the Transpiration Stress Index (TSI), based on the collected temperature and humidity data. This computation helps the system determine the degree of moisture stress that might affect the development of the oyster mushrooms. Using the calculated TSI values, the system can determine if the environment is suitable or if any adjustments need to be made.

The processed data is then stored in the database layer of the system, where the system uses MongoDB as the database for the storage of all the collected environmental data and the calculated TSI values.

Lastly, the system can provide the required information through the application layer of the system, where the farmer can access the real-time data from the sensors and can also access the historical data collected and the required alerts for moisture stress levels that have become critical. The system can also send the required signal for the misting system to provide the required humidity for the mushroom cultivation environment.

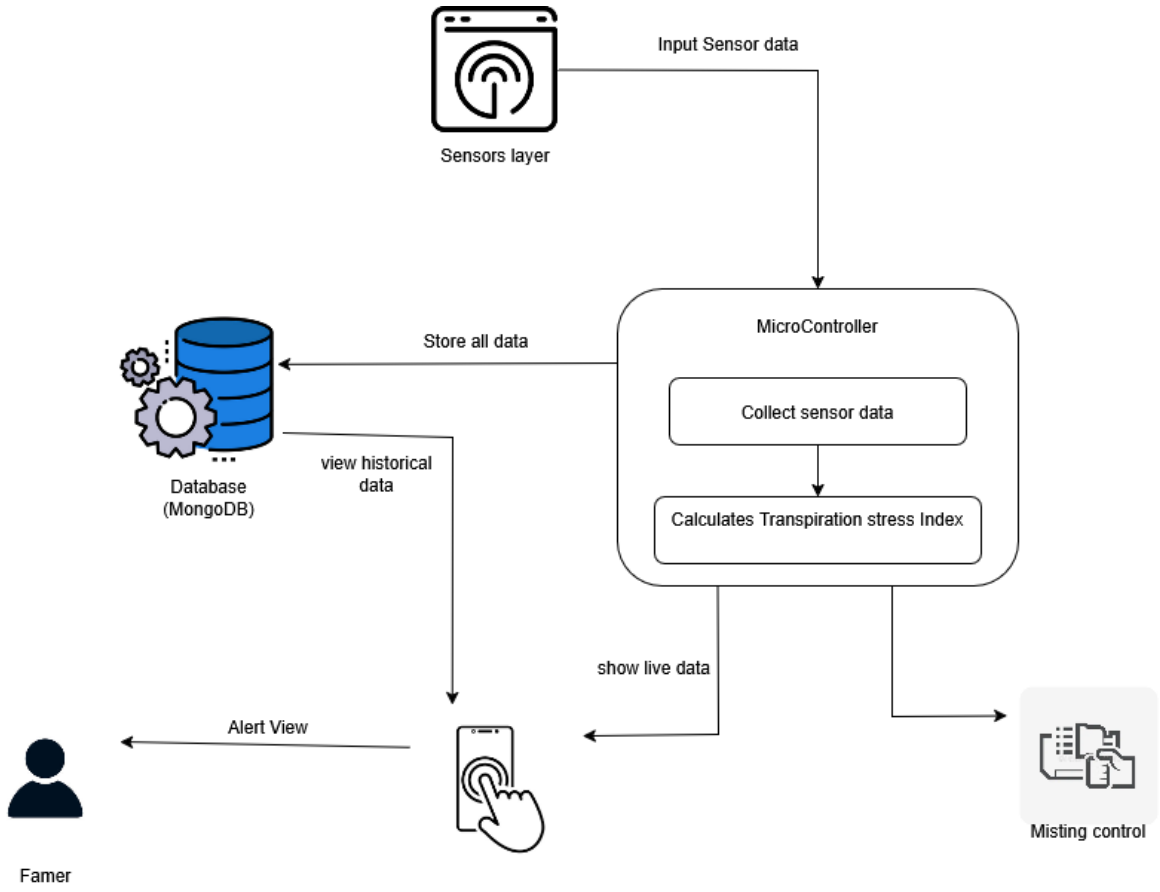


Figure 1.4.1. High level Diagram

5.USER REQUIREMENTS

5.1. Functional Requirements

FR No.	Input	Process	Output
FR-01	Temperature data from the temperature sensor.	The system reads the temperature value and sends it to the monitoring system.	The current temperature is displayed on the system dashboard.
FR-02	Humidity data from the humidity sensor.	The system reads the humidity value and sends it to the system for processing.	The current humidity value is displayed on the dashboard.
FR-03	Temperature and humidity values collected from sensors.	The system stores the collected environmental data in the database.	Environmental data is saved and available for analysis.
FR-04	Temperature and humidity values	The system applies the Transpiration Stress Index (TSI) formula to calculate the stress level.	A calculated TSI value that represents the moisture stress level.
FR-05	Calculated TSI value.	The system compares the TSI value with predefined threshold levels.	The system identifies the condition as: • Low Stress • Moderate Stress • High Stress
FR-06	Environmental data and calculated TSI value	The system analyzes environmental patterns to predict possible moisture stress.	Prediction of future moisture stress conditions.
FR-07	High stress level detected.	The system generates a warning alert	Notification or alert message displayed to the farmer.

Table 5.1:Funtional Requirements

5.2. Non- Functional Requirement

NFR No.	Input	Process	Output
NFR-01	Sensor readings (temperature, humidity)	Real-time data acquisition, Transpiration Stress Index (TSI) calculation	System updates dashboard every 5–10 seconds; low-latency alerts
NFR-02	Continuous sensor data	Data validation and error checking	System operates without crashes; invalid readings are flagged
NFR-03	User requests via dashboard / mobile app	Fetch latest environmental data and historical logs	Dashboard available 24/7; system accessible remotely
NFR-04	Multiple sensor nodes	Aggregate data from all sensors; process for multiple cultivation rooms	Can monitor multiple rooms without performance degradation
NFR-05	Farmer / researcher interaction	Display environmental data, TSI, alerts, and recommendations	Easy-to-read dashboard, clear notifications, simple interface
NFR-06	Login credentials, cloud data	Authenticate user, encrypt data transfers	Prevent unauthorized access; secure data storage
NFR-07	Firmware updates, software patches	Apply system updates without downtime	Easy to update system software and hardware components
NFR-08	Sensor and relay operation	Control energy usage of mist sprayers and pumps	Minimize power consumption while maintaining optimal humidity
NFR-09	Sensor data	Data processing and TSI computation	2°C temperature accuracy,3% humidity accuracy; reliable alerts

Table5.2..Non-Functonal requirement

5.3 Test case

TC No.	Description	Input	Output
TC-01	Check temperature sensor reading	Temperature sensor active	System displays correct temperature value
TC-02	Check humidity sensor reading	Humidity sensor active	System displays correct humidity value
TC-03	Store sensor data	Temperature & humidity data	Data saved in database
TC-04	Calculate TSI value	Temperature and humidity input	System calculates stress index
TC-05	Detect moisture stress	High temperature + low humidity	System shows high stress level
TC-06	Alert generation	Stress level exceeds threshold	Warning alert sent to user
TC-07	Dashboard display	Environmental data available	Dashboard shows temperature, humidity, and stress index
TC-08	Recommendation system	High stress detected	System shows moisture management suggestion

Table 5.3.Test Cases

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The target market for this system will be small and medium scale oyster mushroom farmers who cultivate the product in a controlled or semicontrolled environment. Mushroom cultivation has been gaining popularity in countries such as Sri Lanka because the cultivation process requires relatively less space and can provide the population with an additional source of income.

A small-scale mushroom farm owner or the mushroom growing room owner can be the target customer for this system. Mushroom farmers usually employ manual methods of monitoring the environment and controlling the humidity level in the growing rooms or cultivation houses. However, the environmental conditions may change rapidly in the rooms or houses where the cultivation process takes place. This may cause inconsistent humidity levels in the environment, affecting the growth of the product and the productivity of the farmers. The target customers for this system can be the small-scale mushroom farmers or the owners of the mushroom-growing rooms or small-scale cultivation houses.

Apart from the main users, the secondary users of this system can be the research centers and training institutions in the agricultural sector that need advanced systems for monitoring the environment in order to carry out the cultivation process.

6.2 Value Proposition

The system that is going to be proposed is a smart system for the moisture level in oyster mushroom cultivation**.** The system is different from other monitoring systems that only monitor the humidity and temperature levels individually. Instead, it monitors both levels and computes a transpiration stress index.

The Transpiration Stress Index is a system that helps the farmer understand the real stress levels of the mushroom instead of just looking at the environmental levels. It monitors both temperature and humidity levels and computes them into a single index.

The benefits of using the system include

- Real-time monitoring of environment
- Detection of moisture stress in mushroom crops
- Automated control of moisture through a misting system
- Less human intervention required
- Better quality and productivity of mushrooms

By using this system, farmers can maintain more stable environmental conditions, reduce crop losses, and improve overall farm efficiency.

6.3 Revenue Model

The system can be made to generate revenue through various models. First, the system can be made to generate revenue through the sale of hardware products to the farmers. In this case, the entire monitoring system is sold to the farmer as a package. In such a scenario, the farmer buys a product to be used in his mushroom farming business.

Another possible revenue generation model is through a subscription-based monitoring system. In such a case, the farmer is required to pay a certain amount of money on a monthly basis to use the services offered by the system.

Another possible revenue generation model is through installation services and maintenance support offered to the farmers using the system. This ensures that the farmer is able to use the system in his farming environment.

6.4 Cost Estimation

The estimated hardware cost for developing one prototype system is presented below.

Component	**Estimated Cost (LKR)**
Temperature & Humidity Sensor (DHT22)	2,500
Microcontroller (ESP32 / Arduino)	4,500
Relay Module	1,500
Mist Sprayer / Humidifier	6,000
Water Pump	4,000
Power Supply	2,500
Cables and Connectors	1,500
Enclosure Box	2,000

Total Estimated Cost: Approximately **24,000 – 25,000 LKR per system**.

These costs are based on typical electronic component prices available in local electronics markets in Sri Lanka.

6.5 Pricing Strategy

The system can be sold at a price slightly higher than the production cost to maintain profitability while remaining affordable for farmers.

Example pricing structure:

Estimated production cost: 25,000 LKR

Proposed selling price: 35,000 – 40,000 LKR

This pricing strategy provides a reasonable profit margin while ensuring the product remains accessible to small-scale farmers. Discounts may also be offered for bulk purchases or farmer cooperatives.

6.6 Competitive Advantage

The proposed system has several benefits compared to the current systems for monitoring mushrooms.

Current systems for monitoring mushrooms mainly concentrate on the measurement of environmental factors such as temperature and humidity. However, these systems do not provide clear information regarding the impact of environmental factors on the moisture stress of the mushroom.

The proposed system has been developed based on the calculation of the Transpiration Stress Index (TSI), which measures the combined effect of temperature and humidity factors that cause moisture stress in the mushroom.

In addition, the proposed system has the advantage of being cost-effective and easily deployable in small-scale farming environments compared to the expensive systems that have been deployed in industrial environments.

6.7 Intellectual Property Considerations

The main intellectual property of the system is the algorithm used to compute the Transpiration Stress Index and the combination of environmental monitoring with predictive moisture management.

- To protect this innovation, the following steps can be taken:
- Registering the software system or algorithm as an intellectual property

Making an application for a software copyright and/or patent, as the calculation of the index may result in a new technology

These measures can help prevent unauthorized copying and support the future commercialization of the technology.

7. BUDGET AND JUSTIFICATION

The estimated budget for developing the proposed predictive moisture management system includes hardware components, software tools, cloud services, and development resources. The budget is designed to remain cost-effective while ensuring the development of a functional prototype.

7.1 Hardware Costs

Hardware Component	Quantity	Cost (LKR)
Temperature & Humidity Sensor (DHT22)	1	2,500
Microcontroller (ESP32)	1	4,500
Relay Module	1	1,500
Mist Sprayer / Humidifier	1	6,000
Mini Water Pump	1	4,000
Power Supply Unit	1	2,500
Wires and Connectors	1 set	1,500
Protective Enclosure Box	1	2,000

Total Hardware Cost: 24,000 LKR

Justification:

These hardware components are required to measure environmental conditions and control humidity levels automatically. The sensors collect temperature and humidity data, the microcontroller processes the information and calculates the Transpiration Stress Index, and the relay system activates the misting device to maintain suitable humidity levels.

7.2 Software Costs

Software Tool	Purpose
Arduino IDE / Microcontroller Tools	Sensor data processing and device control
Python Programming Environment	Algorithm development and data analysis
MongoDB Database	Environmental data storage
React Development Tools	Application development

Justification:
Open-source software tools are used for development, which significantly reduces the overall project cost while still providing reliable and flexible development environments.

7.3 Cloud Services

Service	Purpose
Cloud Data Storage	Store environmental sensor data
Cloud Monitoring Dashboard	Remote monitoring and visualization

Justification:
Cloud services allow environmental data to be stored securely and accessed remotely. Farmers can monitor real-time and historical data through a dashboard, which supports better decision-making.

7.4 Total Budget

The estimated total budget combines hardware, software, and cloud service costs required to develop the prototype system.

The budget prioritizes affordable hardware components, open-source software tools, and minimal cloud service costs. This ensures the development of a functional prototype capable of real-time environmental monitoring, Transpiration Stress Index calculation, and automated moisture management, while keeping the system economically feasible for future commercialization.

8.WORK BREAKDOWN STRUCTURE (WBS)

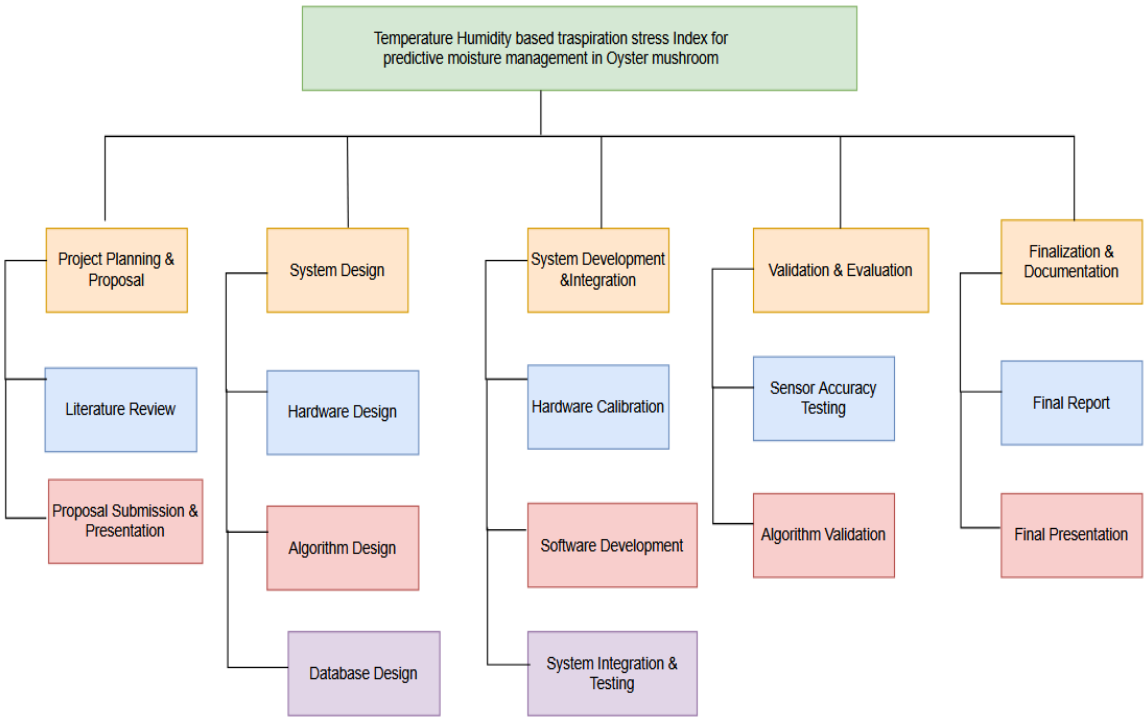


Figure 2.8.WBS Diagram

9.GANTT CHART

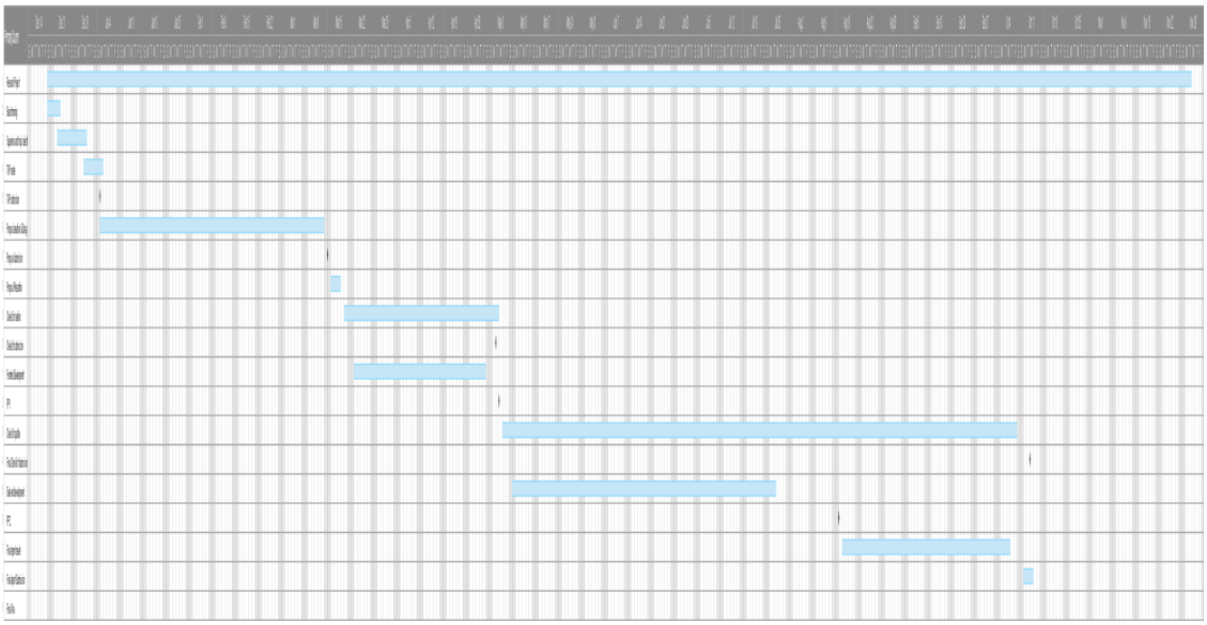


Figure 3.9. Gantt Chatt

https://app.smartsheet.com/sheets/rCQ78fHCFHjh6cg7rFVMp77q274chmmCRjr6Xh21?view=gantt

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APPENDICES

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